

University of Dhaka
Affiliated Engineering Colleges
Department of Computer Science and Engineering
1st Year 1st Semester B.Sc. in CSE Final Examination, 2021
MATH-1105: Differential and Integral Calculus

Total Marks: 70

Time: 3 Hours

(Answer any 5 (Five) of the following Questions)

1. a) Define Continuity and Differentiability of a Function. Page 27, 32 3
b) Test the Continuity and Differentiability of the following function at $x = \frac{\pi}{2}$ 6

$$f(x) = \begin{cases} 1 & \text{when } x < 0 \\ 1 + \sin x & \text{when } 0 \leq x \leq \frac{\pi}{2} \\ 2 + (x - \frac{\pi}{2})^2 & \text{when } x \geq \frac{\pi}{2} \end{cases}$$

- c) Draw the graph of the following function $f(x) = \begin{cases} x - 1; & x > 0 \\ -\frac{1}{2}; & x = 0 \\ x + 1; & x < 0 \end{cases}$ 5

2. a) Explain Exponentially growth and decay with graph. 4
b) Find the Approximation of $f(x) = \sqrt{x}$ at $x = 16$ with graph. 4
c) State L'Hospital Rule. Evaluate (Using L'Hospital rule) 6

i. $\lim_{x \rightarrow 1} (\frac{x}{x-1} - \frac{1}{\ln x})$ ii. $\lim_{x \rightarrow 0} \frac{x - \tan x}{x^3}$

3. a) Write the Walli's formula. Find the reduction formula of $\int \tan^n x dx$ 4
b) Prove that 6

$$\int_0^{\frac{\pi}{2}} \sin^p \theta \cos^q \theta d\theta = \frac{\Gamma(\frac{p+1}{2}) \Gamma(\frac{q+1}{2})}{2\Gamma(\frac{p+q+2}{2})}$$

- c) Evaluate 4

$$\lim_{n \rightarrow \infty} [\frac{1}{n+1} + \frac{1}{n+2} + \dots + \frac{1}{2n}]$$

4. a) Find maximum, minimum value and critical point of the function $f(x) = x^3 - 9x^2 + 15x - 3$ 7
b) If $\ln y = \tan^{-1} x$ then show that $(1+x^2)y_{n+2} + (2nx + 2x - 1)y_{n+1} + n(n+1)y_n = 0$. 7

5. a) Define gamma and beta Function. 2
b) Find the radius of the curvature of the asteroide: $x^{\frac{2}{3}} + y^{\frac{2}{3}} = a^{\frac{2}{3}}$ at any point. 5

- c) Prove that $\beta(m, n) = \frac{\Gamma(m)\Gamma(n)}{\Gamma(m+n)}$ 7

6. a) Evaluate the followings (any two): 3.5x2

(i) $\int_0^1 x^3 \sqrt{1+3x^4} dx$ (ii) $\int_0^{\frac{\pi}{2}} \frac{x \sin x}{1 + \cos^2 x} dx$ (iii) $\int_0^{\frac{\pi}{2}} \frac{\sqrt{\sin x}}{\sqrt{\sin x} + \sqrt{\cos x}} dx$

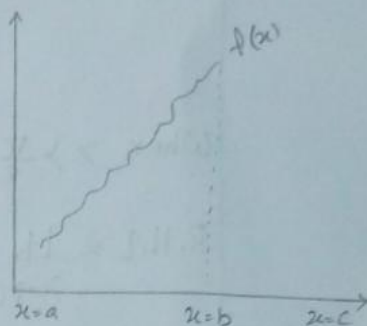
- b) Evaluate the followings (any two): 3.5x2

(i) $\int \frac{1 - \sin x}{x + \cos x} dx$ (ii) $\int \cos^3 x \sin^4 x dx$ (iii) $\int \tan^{-1} x dx$

7. a) Find the area included between the cycloid $x = a(\theta - \sin \theta)$ and $y = a(1 - \cos \theta)$. 8
b) Find the volume formed by revolving the region bounded by the lines $y = x$ and $x = 2$ about X axis. 6

1 a) Continuity of a function:

A function $f(x)$ is said to be continuous at a point $x=a$ if it satisfies the following three condⁿs



i) $f(x)$ is defined at $x=a$

$$\text{ii) } \lim_{x \rightarrow a^-} f(x) = \lim_{x \rightarrow a^+} f(x)$$

$$\text{iii) } f(a) = \lim_{x \rightarrow a} f(x)$$

or $\rightarrow$ Differentiability

Derivative of a function:

$f(x)$ is said to be derivable (or differentiable) at a point if -

$$\lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a} \quad \text{or} \quad \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{\pm h}$$

exists. It is called the derivative or the differential coefficient of the function f at $x=a$ and is denoted by $f'(a)$.

Ayan

b)

$$\text{Given, } f(x) = \begin{cases} 1 & \text{when } x < 0 \\ 1 + \sin x & \text{when } 0 \leq x < \frac{\pi}{2} \\ 2 + (x - \frac{\pi}{2})^2 & \text{when } \frac{\pi}{2} \leq x \end{cases}$$

$$\text{when, } x > \frac{\pi}{2} \text{ then, } f(x) = 2 + (x - \frac{\pi}{2})^2$$

$$\begin{aligned} \text{R.H.L} &= \lim_{x \rightarrow (\frac{\pi}{2})^+} f(x) & \left| \begin{array}{l} \text{Putting,} \\ x = \frac{\pi}{2} + h \end{array} \right. \\ &= \lim_{h \rightarrow 0} f(\frac{\pi}{2} + h) \end{aligned}$$

$$= \lim_{h \rightarrow 0} \left\{ 2 + \left(\frac{\pi}{2} + h - \frac{\pi}{2} \right)^2 \right\}$$

$$= \lim_{h \rightarrow 0} (2 + h^2) = 2$$

$$\text{Again, when, } x < \frac{\pi}{2} \text{ then, } f(x) = 1 + \sin x$$

$$\begin{aligned} \text{L.H.L} &= \lim_{x \rightarrow (\frac{\pi}{2})^-} f(x) & \left| \begin{array}{l} \text{Putting,} \\ x = \frac{\pi}{2} - h \end{array} \right. \\ &= \lim_{h \rightarrow 0} f(\frac{\pi}{2} - h) \end{aligned}$$

$$= \lim_{h \rightarrow 0} \left\{ 1 + \sin(\frac{\pi}{2} - h) \right\}$$

$$= 1 +$$

$$= 1 + \sin\left(\frac{\pi}{2}\right)$$

$$= 1 + 1 = 2$$

Again, when, $x = \frac{\pi}{2}$ then, $f(x) = 2 + (x - \frac{\pi}{2})^2$

$$\therefore f\left(\frac{\pi}{2}\right) = 2 + \left(\frac{\pi}{2} - \frac{\pi}{2}\right)^2$$

$$= 2 = \text{F.V.}$$

Hence, we have,

$$\text{R.H.L} = \text{L.H.L} = \text{F.V.}$$

Therefore, the function $f(x)$ is continuous at $x = \frac{\pi}{2}$. (Ans)

2nd step: Test whether $f(x)$ is differentiable at $x = \frac{\pi}{2}$.

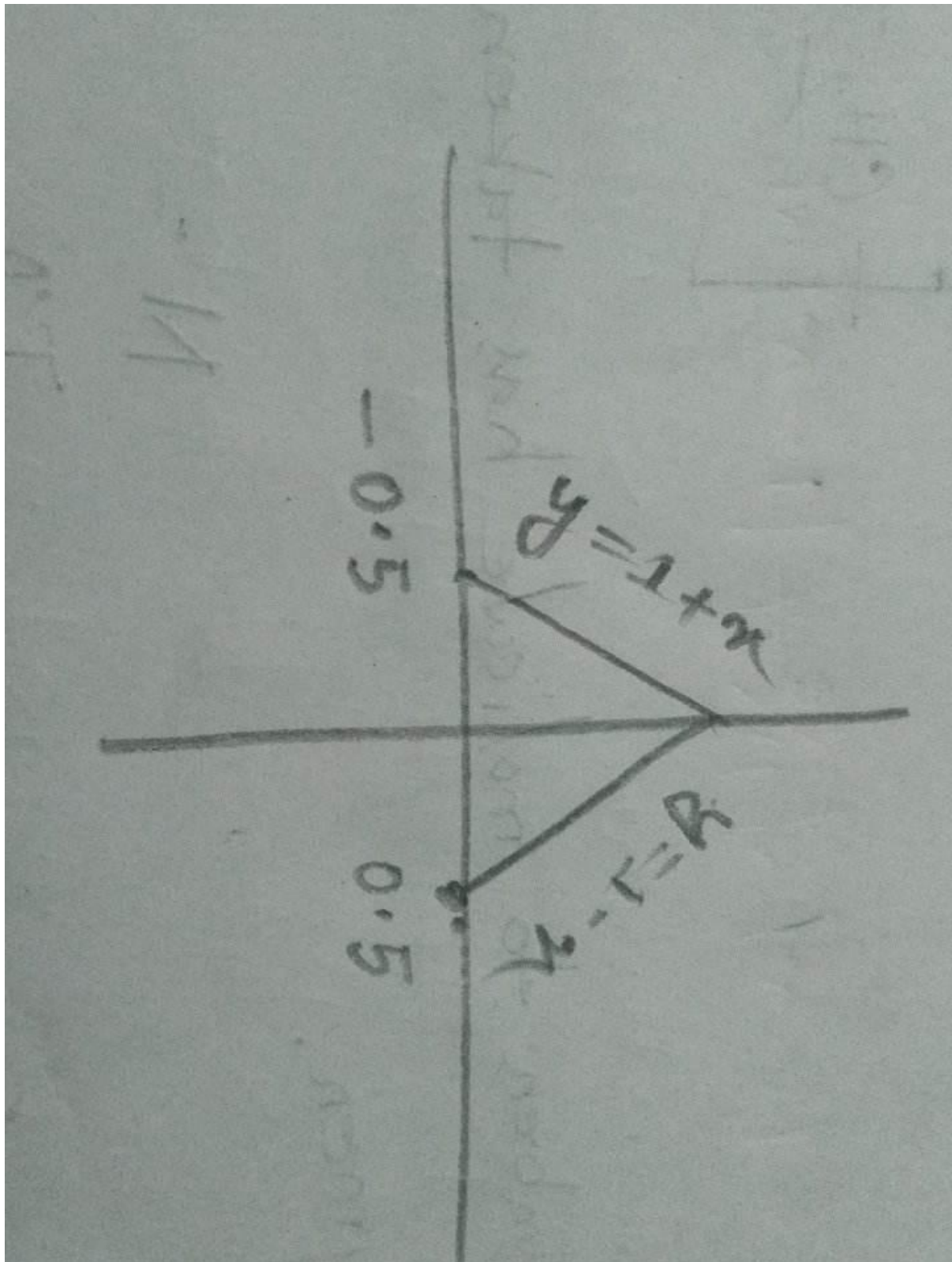
when, $x > \frac{\pi}{2}$ then $f(x) = 2 + (x - \frac{\pi}{2})^2$

$$\begin{aligned}\therefore Rf'(\frac{\pi}{2}) &= \lim_{h \rightarrow 0} \frac{f(\frac{\pi}{2} + h) - f(\frac{\pi}{2})}{h} \\ &= \lim_{h \rightarrow 0} \frac{1}{h} \left[2 + \left(\frac{\pi}{2} + h - \frac{\pi}{2}\right)^2 - 2 + \left(\frac{\pi}{2} - \frac{\pi}{2}\right)^2 \right] \\ &= \lim_{h \rightarrow 0} \frac{1}{h} h^2 = \lim_{h \rightarrow 0} h = 0\end{aligned}$$

Again, $x < \frac{\pi}{2}$ then $f(x) = 1 + \sin x$

$$\begin{aligned}\therefore Lf'(\frac{\pi}{2}) &= \lim_{h \rightarrow 0} \frac{f(\frac{\pi}{2} - h) - f(\frac{\pi}{2})}{-h} \\ &= \lim_{h \rightarrow 0} \frac{1}{-h} \left[1 + \sin(\frac{\pi}{2} - h) - 1 - \sin \frac{\pi}{2} \right] \\ &= \lim_{h \rightarrow 0} \frac{1}{-h} \left[1 + \cos h - 1 - 1 \right] \\ &= \lim_{h \rightarrow 0} \frac{1}{-h} \left[\cos h - 1 \right] \\ &= \lim_{h \rightarrow 0} \frac{1}{-h} \left[1 - \frac{h^2}{2!} + \frac{h^4}{4!} - \dots - 1 \right] \\ &= \lim_{h \rightarrow 0} \frac{1}{-h} h \left[-\frac{h}{2!} + \frac{h^3}{4!} - \dots \right] \\ &= 0\end{aligned}$$

1 (c)



2. (a)

Day

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Sub: _____ Time: _____ Date: / /

Q. Explain exponential growth and decay with graph.

Ans:

Exponential growth

System that exhibits exponential growth according to the mathematical model,

$$y = y_0 e^{kt}$$

where y_0 represents the initial state of the system and $k > 0$ is a constant called the growth constant.

population growth is a common example of exponential growth.

Consider a population of bacteria, for instance.

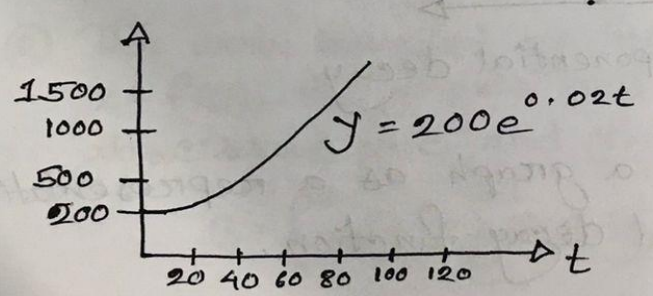


Fig-1: Example of exponential growth of a bacterium.

It seems plausible that the rate of population growth would be proportional to the size of the population. After all, the more bacteria there are to reproduce, the faster the population growth is. Fig-1 represents the growth of population of Bacteria with an initial population of 200 bacterium and a growth constant of 0.02.

Notice that after only (120 minutes) 2 hours, the population is 10 times of its original size.

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Exponential Decay:

Systems that exhibits exponential decay according to the mathematical model,

$$y = y_0 e^{-Kt}$$

where, y_0 represents the initial state of the system and $K > 0$ is a constant is called the decay constant.

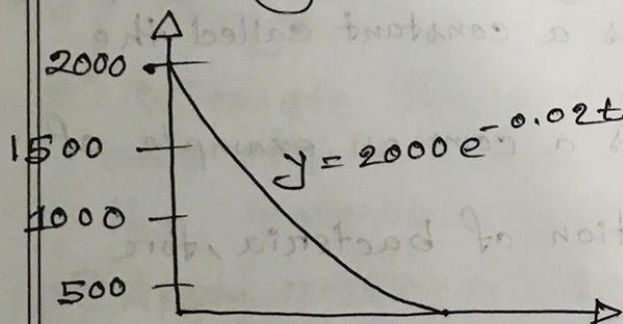


Fig-2: Exponential decay

Fig-2 shows a graph as a representative of exponential decay function.

2.(b)

$$f(x) = \sqrt{x} \quad \text{at } x=16$$

$$\therefore f(16) = \sqrt{16} = 4$$

$$\therefore f'(x) = \frac{1}{2\sqrt{x}} = \frac{1}{2\sqrt{16}} = \frac{1}{8}$$

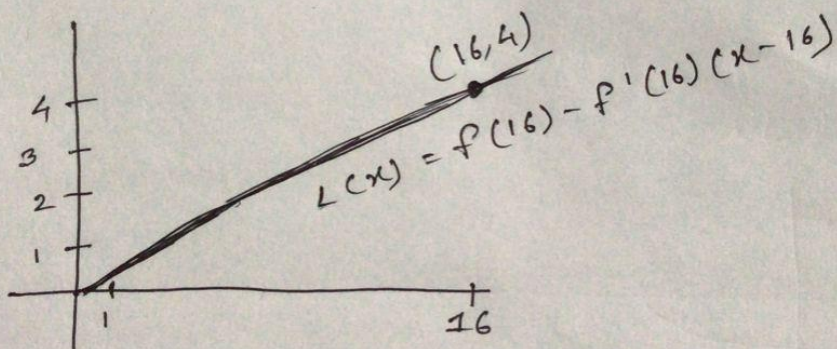
$$\therefore L(x) = f(16) + f'(16)(x-16)$$

$$= 4 + \frac{1}{8}(x-16)$$

$$= 4 + \frac{x-16}{8}$$

$$= \frac{32+x-16}{8}$$

$$= \frac{x}{8} + 2$$



2(c)

$$\begin{aligned} & \lim_{n \rightarrow 1} \left(\frac{n}{n-1} - \frac{1}{\ln n} \right) \quad [(\infty - \infty) \text{ form}] \\ &= \lim_{n \rightarrow 1} \frac{n \ln n - (n-1)}{\ln n (n-1)} \\ &= \lim_{n \rightarrow 1} \frac{n \cdot \frac{1}{n} + \ln n - 1 + 0}{\frac{(n-1)}{n} + \ln n (1-0)} \\ &= \lim_{n \rightarrow 1} \frac{1 + \ln n - 1}{\frac{n-1}{n} + \ln n} \quad \left[\frac{0}{0} \text{ form} \right] \\ &= \lim_{n \rightarrow 1} \frac{\ln n}{\frac{n-1 + n \ln n}{n}} \\ &= \lim_{n \rightarrow 1} \frac{n \ln n}{n-1 + n \ln n} \\ &= \lim_{n \rightarrow 1} \frac{n \cdot \frac{1}{n} + \ln n}{1 - 0 + \ln n + n \cdot \frac{1}{n}} \\ &= \frac{1 + \ln(1)}{1 - 0 + \ln(1) + 1} \\ &= \frac{1 + 0}{1 + 0 + 1} = \frac{1}{2} \quad (\text{Ans.}) \end{aligned}$$

$$\begin{aligned}
 & \lim_{n \rightarrow 0} \frac{n - \tan n}{n^3} \quad \left[\frac{0}{0} \text{ form} \right] \\
 &= \lim_{n \rightarrow 0} \frac{1 - \sec^2 n}{3n^2} = \lim_{n \rightarrow 0} \frac{-2 \sec^2 n \cdot \tan n}{6n} \\
 &= \lim_{n \rightarrow 0} \frac{-2 \sec^4 n - 4 \tan^2 n \sec^2 n}{6} \\
 &= \lim_{n \rightarrow 0} \frac{-2(\sec^4 n + 2 \tan^2 n \sec^2 n)}{6} \\
 &= \lim_{n \rightarrow 0} -\frac{\sec^4 n + 2 \tan^2 n \sec^2 n}{3} \\
 &= -\frac{1+0}{3} \\
 &= -\frac{1}{3} \quad (\text{Ans})
 \end{aligned}$$

Calculus - 2021

3.

a) The wallis formula is,

$$\int_0^{\pi/2} \cos^n x \, dx = \int_0^{\pi/2} \sin^n x \, dx$$

If n is even, $\int_0^{\pi/2} \cos^n x \, dx = \int_0^{\pi/2} \sin^n x \, dx = \frac{1 \cdot 3 \cdot 5 \cdots (n-1)}{2 \cdot 4 \cdot 6 \cdots n} \cdot \frac{\pi}{2}$

If n is odd,

$$\int_0^{\pi/2} \cos^n x \, dx = \int_0^{\pi/2} \sin^n x \, dx = \frac{2 \cdot 4 \cdot 6 \cdots (n-1)}{1 \cdot 3 \cdot 5 \cdots n}$$

Reduction formula of $\int \tan^n x \, dx$,

$$\begin{aligned} \text{Hence, } \int \tan^n x \, dx &= \int (\tan^{n-2} x \cdot \tan^2 x) \, dx \quad \because \tan^2 x = \sec^2 x - 1 \\ &= \int \tan^{n-2} x \cdot (\sec^2 x - 1) \, dx \\ &= \int (\tan^{n-2} x \cdot \sec^2 x - \tan^{n-2} x) \, dx \end{aligned}$$

$$\therefore \int \tan^n x \, dx = \int \tan^{n-2} x \cdot \sec^2 x \, dx - \int \tan^{n-2} x \, dx$$

Let, $u = \tan x$

$$\Rightarrow \frac{du}{dx} = \sec^2 x \quad \Rightarrow du = \sec^2 x \, dx$$

$$\begin{aligned}
 \int \tan^{n-2} x \sec x dx &= \int u^{(n-2)} du \\
 &= \frac{u^{n-2+1}}{n-2+1} + C \\
 &= \frac{u^{n-1}}{n-1} + C
 \end{aligned}$$

Now,

$$\int \tan^n(x) dx = \frac{u^{n-1}}{n-1} + \int \tan^{n-2} x dx + C$$

$$\therefore \int \tan^n x dx = \frac{\tan^{n-1} x}{n-1} + \int \tan^{n-2} x dx + C$$

So the reduction formula of $\int \tan^n x dx$ is,

$$\frac{\tan^{n-1} x}{n-1} + \int \tan^{n-2} x dx + C$$

b)

We know that, $\beta(m, n) = \int_0^1 x^{m-1} \cdot (1-x)^{n-1} dx$

Let, $x = \sin^2 \theta$

$$\Rightarrow 1 = 2 \sin \theta \cos \theta \frac{d\theta}{dx}$$

$$\Rightarrow dx = 2 \sin \theta \cos \theta d\theta$$

x	0	1
θ	0	$\pi/2$

Let, $2m-1 = p$

$$2n-1 = q$$

and, $m = \frac{p+1}{2}$

$$n = \frac{q+1}{2}$$

Now,

$$\beta(m, n) = \int_0^{\pi/2} \sin^{2(m-1)-1} \theta \cdot (1 - \sin^2 \theta)^{n-1} \cdot 2 \sin \theta \cos \theta d\theta$$

$$\Rightarrow \beta(m, n) = 2 \int_0^{\pi/2} \sin^{2m-2} \theta \cdot \cos^{2n-2} \theta \cdot \sin \theta \cdot \cos \theta \cdot d\theta$$

$$\Rightarrow \beta(m, n) = 2 \int_0^{\pi/2} \sin^{2m-2+1} \theta \cdot \cos^{2n-2+1} \theta d\theta$$

$$\Rightarrow \beta(m, n) = 2 \int_0^{\pi/2} \sin^{2m-1} \theta \cdot \cos^{2n-1} \theta d\theta$$

$$\Rightarrow \beta\left(\frac{p+1}{2}, \frac{q+1}{2}\right) = 2 \int_0^{\pi/2} \sin^p \theta \cdot \cos^q \theta d\theta$$

$$\Rightarrow \frac{\left[\frac{p+1}{2} \right] \left[\frac{q+1}{2} \right]}{\left[\frac{p+1}{2} + \frac{q+1}{2} \right]} = 2 \int_0^{\pi/2} \sin^p \theta \cos^q \theta d\theta$$

$$\therefore \int \sin^p \theta \cos^q \theta d\theta = \frac{\left[\frac{p+1}{2} \right] \left[\frac{q+1}{2} \right]}{2 \left[\frac{p+q+2}{2} \right]} \quad (\text{Proved})$$

c)

Here, Let,

$$\lim_{n \rightarrow \infty} \left[\frac{1}{n+1} + \frac{1}{n+2} + \dots + \frac{1}{2n} \right] = P$$

$$\Rightarrow P = \lim_{n \rightarrow \infty} \sum_{r=1}^n \frac{1}{n+r}$$

$$\Rightarrow P = \lim_{n \rightarrow \infty} \sum_{r=1}^n \frac{1}{n} \cdot \frac{1}{\left(1 + \frac{r}{n}\right)}$$

$$\Rightarrow P = \int_0^1 \frac{dx}{1+x}$$

$$\Rightarrow P = \left[\log(1+x) \right]_0^1 = \log 2 - \log 1 = \log 2$$

$$\therefore \lim_{n \rightarrow \infty} \left[\frac{1}{n+1} + \frac{1}{n+2} + \dots + \frac{1}{2n} \right] = \log 2$$

putting $\frac{r}{n} = x$
and $\frac{1}{n} = dx$
and for $r=0$
 $\frac{r}{n} = 0$
and $r=n$,
 $\frac{r}{n} = 1$

c) Hence,

$$\lim_{n \rightarrow \infty} \left[\frac{1}{n+1} + \frac{1}{n+2} + \dots + \frac{1}{2n} \right]$$
$$\Rightarrow T_{n+1} = \frac{1}{n+1} = \frac{1}{n} \cdot \frac{1}{\left(1 + \frac{r}{n}\right)} \quad \left[\begin{array}{l} \text{putting } \frac{r}{n} = x \\ \text{and } \frac{1}{n} = dx \end{array} \right]$$
$$= \frac{1}{1+x} dx$$

Now, ~~for~~ for the first term $r=0$ and last term $r=n$

$\therefore$ The lower limit of integration $= \lim_{n \rightarrow \infty} \left(\frac{0}{n} \right) = 0$

and the upper limit of integration $= \lim_{n \rightarrow \infty} \left(\frac{n}{n} \right) = 1$

Hence, the required limit,

$$= \int_0^1 \frac{dx}{1+x}$$

$$= \left[\log(1+x) \right]_0^1$$

$$= \log 2 - \log 1$$

$$= \log 2 \quad (\text{Ans.})$$

④

a)

$$f(u) = u^3 - 9u^2 + 15u - 3$$

$$f'(u) = 3u^2 - 18u + 15$$

$$f''(u) = 6u - 18$$

for a minimum or maximum value of $f(u)$,

$$f'(u) = 0$$

$$3u^2 - 18u + 15 = 0$$

$$\Rightarrow 3u^2 - 15u - 3u + 15 = 0$$

$$\Rightarrow (3u - 3)(u - 5) = 0$$

$$\therefore u = 5, 1$$

the critical point of the function is for $u = 5$, $u = 5, 1$.

$$f''(5) = 6 \times 5 - 18$$

$$= 30 - 18$$

$$= 12$$

$$= +ve$$

$f(u)$ is minimum for $u = 5$.

the minimum value of

$$f(u) = 5^3 - 2 \times 5^2 + 15 \times 5 - 3$$
$$= -28$$

Again for $x=1$, ~~$f''(x)$~~

$$f''(1) = 6 \times 1 - 18$$

$$= -12$$

$f(0)$ is maximum for $x=1$.

the maximum value of

$f(x)$ is,

$$f(1) = 1^3 - 2 \times 1^2 + 15 \times 1 - 3$$
$$= 4$$

Therefore the minimum value of the function is -28 & the maximum value of the function is 4. The critical point of the function is 5 & 1.

⑥ given that,

$$\ln y_2 = \tan^{-1} u$$

$$\Rightarrow y_2 = e^{\tan^{-1} u}$$

$$\Rightarrow y_2 = e^{\tan^{-1} u} \cdot \frac{1}{1+u^2}$$

$$\Rightarrow y_1 (1+u^2) = y_2$$

$$\Rightarrow y_2 (1+u^2) + 2u y_1 = y_1$$

$$\Rightarrow y_2 (1+u^2) + 2u y_1 - y_1 = 0$$

differentiate the function n times.

$$\Rightarrow (1+u^2) y_{n+2} + y_{n+1} (2nu + 2n - 1)$$

$$+ (n^2 + n) y_n = 0$$

$$\Rightarrow (1+u^2) y_{n+2} + (2nu + 2n - 1) y_{n+1}$$

$$+ n(n+1) y_n = 0$$

For maximum value of

the function is $\frac{1}{2}$.

the value of the function is $\frac{1}{2}$.

05. (a)

Gamma function:

Gamma function is an extension of the factorial function and is denoted by Γ and defined by, Improper integral

$$\Gamma n = \int_0^{\infty} e^{-x} \cdot x^{n-1} dx$$

Beta function: Definite integral,

$$\int_0^1 x^{m-1} (1-x)^{n-1} dx ; m > 0, n > 0$$

is called Beta function. It is denoted

$$\text{by } B(m, n) = \int_0^1 x^{m-1} (1-x)^{n-1} dx ; m > 0, n > 0$$

5(b)

Given,

the asteroid,

$$x^{2/3} + y^{2/3} = a^{2/3}$$

let, $\boxed{x = a^3 \cos^3 \theta}$, $\boxed{y = a^3 \sin^3 \theta}$

$$\Rightarrow \frac{dx}{d\theta} = a^3 3 \cos^2 \theta \cdot (-\sin \theta)$$

And, $\frac{dy}{d\theta} = a^3 3 \sin^2 \theta (\cos \theta)$

$$\therefore dx = a^3 3 \cos^2 \theta \cdot (-\sin \theta) d\theta, \quad dy = a^3 3 \sin^2 \theta (\cos \theta) d\theta$$

~~$\frac{dy}{dx} = \frac{a^3 3 \sin^2 \theta (\cos \theta) d\theta}{a^3 3 \cos^2 \theta (-\sin \theta) d\theta}$~~

$$\therefore \frac{dy}{dx} = \frac{a^3 3 \sin^2 \theta (\cos \theta) d\theta}{a^3 3 \cos^2 \theta (-\sin \theta) d\theta}$$
$$= \frac{\sin \theta}{-\cos \theta} = -\tan \theta$$

$$\Rightarrow \frac{d^2 y}{dx^2} = -\sec^2 \theta \cdot \frac{d\theta}{dx}$$
$$= -\sec^2 \theta \cdot \frac{1}{-a^3 3 \cos^2 \theta \sin \theta}$$
$$= \frac{-\sec^2 \theta}{-a^3 3 \cos^2 \theta \sin \theta} = \frac{\frac{1}{\cos^2 \theta}}{a^3 3 \cos^2 \theta \sin \theta}$$
$$= \frac{1}{3a^3 \cos^4 \theta \cdot \sin \theta}$$

we know,

$$\text{Radius of a curvature at } (x, y) \text{ point, } R = \frac{\left(1 + \left(\frac{dy}{dx}\right)^2\right)^{3/2}}{\left|\frac{d^2y}{dx^2}\right|}$$

$$= \frac{(1 + \tan^2 \theta)^{3/2}}{d^2y/dx^2} \quad ; \quad \left[\frac{dy}{dx} = -\tan \theta \right]$$

$$= \frac{(1 + \tan^2 \theta)^{3/2}}{1/3a^3 \cos^4 \theta \cdot \sin \theta}$$

$$= \frac{(\sec^2 \theta)^{3/2}}{1}$$
$$= \frac{\sec^3 \theta}{3a^3 \cos^4 \theta \sin \theta}$$

$$= \frac{\sec^3 \theta}{1}$$

$$= 3a^3 \sec^3 \theta \times 3a^3 \cos^4 \theta \sin \theta$$

$$= 3a^3 \cos \theta \sin \theta$$

$$= \frac{3a^3 (xy)^{1/3}}{a^2}$$

$$= 3a (xy)^{1/3}$$

(Ans.)

$$x = a^3 \cos^3 \theta$$
$$\Rightarrow \cos \theta = \frac{\sqrt[3]{x}}{a}$$

$$y = a^3 \sin^3 \theta$$
$$\Rightarrow \sin \theta = \frac{\sqrt[3]{y}}{a}$$

$$\therefore \cos \theta \sin \theta = \frac{x^{1/3} \cdot y^{1/3}}{a^2}$$
$$= \frac{(xy)^{1/3}}{a^2}$$

5(c) (2024) $B(m, n) = \frac{\Gamma(m) \cdot \Gamma(n)}{\Gamma(m+n)}$ Prove

We know,

$$\Gamma(m) = \int_0^{\infty} e^{-x} x^{m-1} dx \quad \text{--- (1)}$$

let,
 $x = \lambda y$

$$\Rightarrow 1 = \lambda \frac{dy}{dx}$$

$$\Rightarrow dx = \lambda dy$$

From (1) we can write,

$$\begin{aligned} \Gamma(m) &= \int_0^{\infty} e^{-\lambda y} (\lambda y)^{m-1} \cdot \lambda dy \\ &= \int_0^{\infty} e^{-\lambda y} \cdot \lambda^m \cdot y^{m-1} \cdot dy \quad \text{--- (ii)} \end{aligned}$$

Now we can write,

~~$$\Gamma(n) = \int_0^{\infty} e^{-\lambda} \cdot \lambda^{n-1} dy$$~~

$$\boxed{\Gamma(n) = \int_0^{\infty} e^{-\lambda} \cdot \lambda^{n-1} dy \quad \text{--- (iii)}}$$

Multiplying (2) and (3)

$$\begin{aligned} \Gamma(m) \Gamma(n) &= \int_0^{\infty} \int_0^{\infty} e^{-\lambda y - \lambda} \cdot \lambda^{m+n-1} \cdot y^{m-1} dy d\lambda \\ &= \int_0^{\infty} \left[\int_0^{\infty} e^{-\lambda(1+y)} \lambda^{m+n-1} d\lambda \right] y^{m-1} dy \end{aligned}$$

$$\Rightarrow \Gamma(m) \Gamma(n) = \int_0^{\infty} \frac{\Gamma(m+n)}{(1+y)^{m+n}} y^{m-1} dy$$

$$; \boxed{\int_0^{\infty} e^{-kt} \cdot t^{n-1} dt = \frac{\Gamma(n)}{k^n}}$$

$$\Rightarrow \Gamma(m) \Gamma(n) = \Gamma(m+n) \int_0^{\infty} \frac{y^{m-1}}{(1+y)^{m+n}} dy$$

$$\Rightarrow \Gamma(m) \cdot \Gamma(n) = \Gamma(m+n) \cdot B(m, n); \boxed{B(m, n) = \int_0^{\infty} \frac{y^{n-1}}{(1+y)^{m+n}} dy}$$

$$\Rightarrow B(m, n) = \frac{\Gamma(m) \Gamma(n)}{\Gamma(m+n)}$$

(Proved)

6(a)

Date: / /

$$\textcircled{1} \int_0^1 x^3 \sqrt{1+3x^4} \, dx$$

$$\text{let } 1+3x^4 = z^2$$

diff. both side w.r.t x

$$\int_1^2 \frac{1}{6} z \, dz$$

$$0 + 3 \cdot 4x^3 \, dx = 2z \, dz$$

$$\Rightarrow \frac{1}{6} \int_1^2 z^2 \, dz$$

$$+2x^3$$

$$12x^3 \, dx = 2z \, dz$$

$$\Rightarrow \frac{1}{6} \left[\frac{z^3}{3} \right]_1^2$$

$$\int x^n \, dx = \frac{x^{n+1}}{n+1}$$

$$x^3 \, dx = \frac{z}{12} \, dz$$

$$= \frac{1}{6} \left[\frac{2^3}{3} - \frac{1^3}{3} \right]$$

x	0	1
z	1	2

$$= \frac{1}{6} \times \left[\frac{8}{3} - \frac{1}{3} \right]$$

$$= \frac{1}{6} \times \frac{7}{3}$$

$$= \frac{7}{18} \text{ (ans)}$$

6. a. ii

Date: / /

$$\int_0^{\pi} \frac{x \sin x}{1 + \cos^2 x} dx$$

$$\int_0^a f(x) dx = \int_0^a f(a-x) dx$$

$$\text{Let } I = \int_0^{\pi} \frac{x \sin x}{1 + \cos^2 x} dx \quad \text{--- (i)}$$

$$I = \int_0^{\pi} \frac{(\pi-x) \sin(\pi-x)}{1 + (\cos(\pi-x))^2} dx$$

$$I = \int_0^{\pi} \frac{(\pi-x) \sin x}{1 + \cos^2 x} dx$$

$$I = \int_0^{\pi} \left(\frac{\pi \sin x}{1 + \cos^2 x} - \frac{x \sin x}{1 + \cos^2 x} \right) dx$$

$$I = \int_0^{\pi} \frac{\pi \sin x}{1 + \cos^2 x} dx - \int_0^{\pi} \frac{x \sin x}{1 + \cos^2 x} dx$$

$$I = \int_0^{\pi} \frac{\pi \sin x}{1 + \cos^2 x} dx - \int_0^{\pi} \frac{x \sin x}{1 + \cos^2 x} dx \quad \text{--- (ii)}$$

$$\text{(i) + (ii) } \Rightarrow$$

$$2I = \pi \int_0^{\pi} \frac{\sin x}{1 + \cos^2 x} dx$$

$$2I = \pi \int_{-1}^1 \frac{1}{1+t^2} (-) dt \quad \text{put } \cos u = t$$

$$- \sin u du = dt$$

$$2I = \pi \int_{-1}^1 \frac{1}{1+t^2} dt$$

$$2I = \pi \left[\tan^{-1} t \right]_{-1}^1$$

$$I = \pi/2 \left[\tan^{-1}(1) - \tan^{-1}(-1) \right]$$

$$= \pi/2 \left[\tan^{-1}(1) - (-\tan^{-1} 1) \right]$$

$$= \pi/2 \left[\tan^{-1}(1) + \tan^{-1} 1 \right]$$

$$= \pi/2 \left[2 \tan^{-1}(1) \right]$$

$$= \pi \tan^{-1}(1)$$

$$= \pi \times \pi/4 = \frac{\pi^2}{4} \text{ (ans)}$$

$$\begin{aligned} \tan^{-1}(-u) \\ = -\tan^{-1} u \end{aligned}$$

6. a. iii

Integral calculus:

$$1. \text{ let } I = \int_0^{\pi/2} \frac{\sqrt{\sin u}}{\sqrt{\sin u} + \sqrt{\cos u}} \quad \text{--- (i)}$$

$$\int_0^a f(u) du = \int_0^a f(a-u) du$$

$$= \int_0^{\pi/2} \frac{\sqrt{\sin(\pi/2 - u)}}{\sqrt{\sin(\pi/2 - u)} + \sqrt{\cos(\pi/2 - u)}} (du)$$

$$I = \int_0^{\pi/2} \frac{\sqrt{\cos u}}{\sqrt{\cos u} + \sqrt{\sin u}} du \quad \text{--- (ii)}$$

Now (i) + (ii) $\Rightarrow$

$$I + I = \int_0^{\pi/2} \frac{\sqrt{\sin u}}{\sqrt{\sin u} + \sqrt{\cos u}} du + \int_0^{\pi/2} \frac{\sqrt{\cos u}}{\sqrt{\sin u} + \sqrt{\cos u}} du$$

$$= \int_0^{\pi/2} \left(\frac{\sqrt{\sin u} + \sqrt{\cos u}}{\sqrt{\sin u} + \sqrt{\cos u}} \right) du$$

$$= [u]_0^{\pi/2}$$

$$2I = \pi/2 - 0$$

$$\Rightarrow I = \pi/4$$

② $\pi/2$

$$\int_0^{\pi/2} \frac{du}{1 + \cot u}$$

$$\text{Let } I = \int_0^{\pi/2} \frac{1}{1 + \cot u} du = \int_0^{\pi/2} \frac{1}{1 + \frac{\cos u}{\sin u}} du$$

$$I = \int_0^{\pi/2} \frac{1}{\frac{\sin u + \cos u}{\sin u}} du = \int_0^{\pi/2} \frac{\sin u}{\sin u + \cos u} du \quad \text{--- (i)}$$

$$I = \int_0^{\pi/2} \frac{\sin(\pi/2 - u)}{\sin(\pi/2 - u) + \cos(\pi/2 - u)} du \quad \left(\int_0^a f(u) du = \int_0^a f(a-u) du \right)$$

$$I = \int_0^{\pi/2} \frac{\cos u}{\cos u + \sin u} du \quad \text{--- (ii)}$$

Adding (i) and (ii), we get

$$2I = \int_0^{\pi/2} \left(\frac{\sin u}{\sin u + \cos u} + \frac{\cos u}{\sin u + \cos u} \right) du$$

$$2I = \int_0^{\pi/2} \left(\frac{\cancel{\sin u} + \cancel{\cos u}}{\cancel{\sin u} + \cancel{\cos u}} \right) du$$

$$2I = \int_0^{\pi/2} du = [u]_0^{\pi/2} = \pi/2 - 0$$

$$2I = \pi/2$$

$$\therefore I = \pi/4$$

6. (b)

$$\textcircled{1} \int \frac{1 - \sin x}{x + \cos x} dx.$$

$$= \int \frac{dt}{t}.$$

$$= \ln |t| + c$$

$$= \ln (|x + \cos x|) + c; c \in \mathbb{R}$$

A

$$\begin{array}{l} \text{let,} \\ x + \cos x = t \\ (1 - \sin x) dx = dt \end{array}$$

6. b

$$\begin{aligned} \textcircled{ii} \quad & \int \cos^5 x \sin^4 x \, dx \\ &= \int \sin^4 x \cos^4 x \cdot \cos x \, dx. \quad \left| \begin{array}{l} \sin x = t \\ \cos x \, dx = dt \end{array} \right. \\ &= \int \sin^4 x (1 - \sin^2 x) \cos x \cdot (1 - \sin^2 x) \, dx \\ &= \int \sin^4 x (1 - \sin^2 x) \cos x \cdot (1 - \sin^2 x) \, dx \\ &= \int t^4 (1 - t^2) (1 - t^2) \cdot dt \\ &= \int t^4 - 2t^6 + t^8 \, dt \\ &= \int t^4 \, dt - \int 2t^6 \, dt + \int t^8 \, dt \\ &= \frac{t^5}{5} - \frac{2t^7}{7} + \frac{t^9}{9} \\ &= \frac{\sin^5 x}{5} - \frac{2 \sin^7 x}{7} + \frac{\sin^9 x}{9} \quad \text{A-} \end{aligned}$$

$$\text{iii)} \quad \tan^{-1} x \, dx$$

$$= \int \textcircled{1} = \tan^{-1} x \int dx - \int \left(\frac{d}{dx} (\tan^{-1} x) \right) \int dx \quad \text{by using the formula}$$

$$= x \tan^{-1} x \, dx - \int \frac{x}{1+x^2} \cdot dx \quad \left[\frac{d}{dx} \tan^{-1} x = \frac{1}{1+x^2} \right]$$

= Now,

$$\int \frac{x}{1+x^2} \, dx$$

$$= \int \frac{\frac{1}{2} dt}{t}$$

$$= \frac{1}{2} \int \frac{1}{t} \, dt$$

$$= \frac{1}{2} \ln |t| + c \quad \left[\int \frac{1}{t} \, dt = \ln |t| + c \right]$$

$$= \frac{1}{2} \ln(1+x^2) + c$$

From equation no $\textcircled{1}$ -

$$x \tan^{-1} x \, dx - \int \frac{x}{1+x^2} \, dx$$

$$\therefore x \tan^{-1} x \, dx - \frac{1}{2} \ln(1+x^2) + c \quad \text{Ans}$$

7.

a) Solⁿ:

Given equations are,

$$x = a(\theta - \sin\theta), \quad y = a(1 - \cos\theta)$$

$$\Rightarrow \frac{dx}{d\theta} = a(1 - \cos\theta)$$

Therefore, the required area of the region

= area of OAPQ

$$= \int_0^{2\pi} y \frac{dx}{d\theta} d\theta$$

$$= \int_0^{2\pi} a(1 - \cos\theta) \cdot a(1 - \cos\theta) d\theta$$

$$= \int_0^{2\pi} a^2 (1 - \cos\theta)^2 d\theta$$

$$= a^2 \int_0^{2\pi} \left(2\sin^2 \frac{\theta}{2}\right)^2 d\theta$$

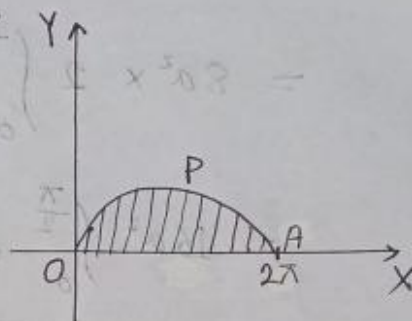
$$= 4a^2 \int_0^{2\pi} \sin^4 \frac{\theta}{2} d\theta$$

Let,

$$\frac{\theta}{2} = \phi$$

$$\Rightarrow \frac{d\theta}{2d\phi} = 1$$

$$\Rightarrow d\theta = 2d\phi$$



$$= 4a^2 \int_0^{\pi} \sin^4 \phi \cdot 2 d\phi$$

$$= 8a^2 \int_0^{\pi} \sin^4 \phi \cdot d\phi$$

$$= 8a^2 \times 2 \int_0^{\frac{\pi}{2}} \sin^4 \phi \cdot d\phi$$

$$= 16a^2 \int_0^{\frac{\pi}{2}} \sin^4 \phi \cdot \cos^0 \phi \cdot d\phi$$

$$= 16a^2 \frac{\sqrt{\frac{4+1}{2}} \sqrt{\frac{0+1}{2}}}{2 \sqrt{\frac{4+0+2}{2}}}$$

$$= 16a^2 \frac{\sqrt{\frac{5}{2}} \sqrt{\frac{1}{2}}}{2 \sqrt{\frac{6}{2}}}$$

$$= 16a^2 \frac{\sqrt{1+\frac{3}{2}} \sqrt{\frac{1}{2}}}{2 \cdot \sqrt{3}}$$

$$= 8a^2 \frac{\frac{3}{2} \sqrt{\frac{3}{2}} \sqrt{\frac{1}{2}}}{2!}$$

$$= 8a^2 \frac{\frac{3}{2} \sqrt{1+\frac{1}{2}} \sqrt{\frac{1}{2}}}{2 \times 1!}$$

θ	0	2π
ϕ	0	π

Formula:

$$\int_0^{\pi} \sin \theta \cdot d\theta$$

$$= 2 \int_0^{\frac{\pi}{2}} \sin \theta \cdot d\theta$$

$$\begin{aligned}
 &= 4a^2 \frac{3}{2} \times \frac{1}{2} \left[\frac{1}{2} \right] \left[\frac{1}{2} \right] \\
 &= 3a^2 \sqrt{\pi} \sqrt{\pi} \quad \left| \sqrt{\pi} = \left[\frac{1}{2} \right] \right. \\
 &= 3a^2 \pi \text{ unit}^2 \text{ (Ans:)}
 \end{aligned}$$

b) Solⁿ: (12/10/19)

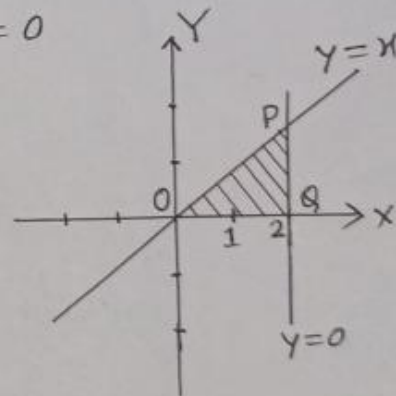
Given lines,

$y = x$ and $x = 2$ and X -axis.

We know, at x -axis, $y = 0$

$$\therefore y = 0$$

$$\Rightarrow x = 0$$



Therefore the required area of the region

= area of $OPQO$

$$= \int_0^2 y \, dx \quad [O(0,0) \text{ and } Q(2,0)]$$

$$= \int_0^2 x \, dx \quad [y=x \text{ given}]$$

$$= \left[\frac{x^2}{2} \right]_0^2$$

$$= \frac{2^2}{2} - \frac{0^2}{2}$$

$$= \frac{4}{2} = 2 \text{ unit}^2 \quad (\underline{\underline{\text{Ans}}})$$

